Global Futures Fund 1.0 (Pilot) - Letter of Interest (LOI)

1. Initiative title

Creating Canada's Largest Capstone Ecosystem

2. Existing or new

Existing

3. Lead applicant

Co-PIs:

- Derek Rayside, Associate Dean, Co-operative Education & Professional Affairs,
- Michael Wood, Associate Dean of Work-Integrated Learning, Faculty of Environment
- Jennifer Yessis, Associate Director, Professional Graduate Studies, School of Public Health Sciences

4. Team members

*Name and position at University of Waterloo. *For external members, list name, position, and organization.*

This proposal is supported by at least one associate dean from each faculty. There is a mixture of associate dean undergraduate, graduate, and co-op/WIL portfolios.

Team Members (in alphabetical order):

- **Associate Deans**
  - Kathy Acheson, Associate Dean, Undergraduate Programs, Faculty of Arts
  - Anna Esselment, Associate Dean, Graduate Studies, Faculty of Arts
  - Leeann Ferries, Associate Dean, Undergraduate Studies, Faculty of Health
  - Bradley Pomeroy, Associate Dean, Graduate Studies, Faculty of Arts (incoming)
  - Mary Robinson, Associate Dean Outreach, Equity, and Diversity, Engineering
- **Curricular Partners**
  - ARTS450 Global Engagement Seminar (Doug Peers, Arts)
  - Wicked Problems (Anna Esselment, Arts; Kyle Scholz CTE)
  - SLICCs (Katherine Lithgow, CTE)
- **Organizational Partners**
  - GreenHouse (Tania Del Matto)
  - Velocity (John Dick & Ben Graham)

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    - Future Cities Institute (Leia Minaker & Eryn Stewart)
    - Centre for Work-Integrated Learning (Andrea Prier)
    - RoboHub (William Melek & Brandon DeHart)
    - Ideas Clinic (Simar Saini & Chris Rennick)
    - Grebel Peace Incubator (Paul Heidebrecht)
- **Other Partners:**
    - Sean Geobey, Associate Professor, School of Environment, Enterprise and Development, Faculty of Environment
    - Rob Gorbet, Associate Professor, Knowledge Integration, Faculty of Environment
    - Diane Williams, Associate Director Undergraduate Studies, School of Public Health Sciences, Faculty of Health
    - Brendan Wylie-Toal, School of Environment, Enterprise and Development, Faculty of Environment

5. Executive sponsor(s)

Enter the name and Faculty(ies) of the Dean(s) who will sponsor this initiative.

We are seeking endorsement from all six deans.

- Arts - Dean Tcheuyap
- Engineering – Dean Wells
- Environment - Dean Frayne
- Health – Dean Liu
- Math – Dean Giesbrecht
- Science – Dean Houser

6. Funding requested

Enter total amount in Canadian dollars.

$250,000 x 3 years = ***$750,000***

**If you proceed to the full proposal, a detailed budget will be required.*

7.Proposed duration of initiative

3 years

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8. Identify one or more aspects of the University's mission that this initiative align(s) with

[] Research/scholarship

[] Education

[] Service

[next page if all filled in]

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Short Responses

Please provide short responses to the following questions. Please keep to an overall maximum of 2,000 words across all responses.

Briefly describe the initiative and how it impacts one or more of the Global Futures.

UW is the world's largest co-op program, and already has the foundations to become Canada's largest capstone ecosystem. Achieving that goal will require coordinating our existing efforts and expertise, and strategic growth in three complementary directions.

UW currently graduates over 2,000 students per year with a mandatory capstone degree requirement, across four faculties: Arts, Environment, Health, and Engineering. Those 2,000+ students complete 500+ capstone projects per year. There are an additional ~6,000 students who currently graduate each year without a capstone opportunity. Currently, perhaps ~5% of these 500+ capstone projects have meaningful connections with external partners.

To become Canada's largest capstone ecosystem, and realize the One Waterloo vision, UW needs to grow strategically in three complementary directions: (1) elective capstone opportunities for the ~6,000 students who currently graduate without such an opportunity; (2) apply the expertise we have developed running co-op at scale to running capstone Work-Integrated Learning at scale; and (3) similarly apply that expertise to grow the Community-Embedded Classroom (CEC) model.

The five-year targets are ~3,000 students per year in a capstone course, with ~300 capstone projects per year engaged with external partners, and ~500 students per year in a CEC course (which could be a capstone course or a pre-capstone course). In the three years of this proposal the targets are 80% of those 5-year targets: ~2,400 capstone students per year, ~240 capstone projects with external partners, and ~400 students in CEC courses.

These thousands of students and hundreds of projects will contribute to all five of the Global Futures. This proposal includes Global Futures Capstone Awards, amongst other aspects. New capstone sections can be explicitly organized around Global Futures themes. *In a sense, this proposal is creating Global Futures Networks for undergraduates and UW's community partners.*

UW is currently behind our peer universities in terms of capstone engagement with community partners. Many of our peers (e.g., McGill, Queens, Toronto, UBC, etc.) already have developed capstone external partner programs in Engineering. UW can leapfrog the competition, because we already have larger foundations: we just need to build the superstructure and open the doors. UW can be the first in Canada to do this at the university level.

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Describe how the initiative leverages existing and/or emerging academic and/or research strengths at UWaterloo. Describe, where applicable, how our founding differentiator strengths are being leveraged.

As the global leader in co-operative education, UW is uniquely positioned to build on this foundation to scale work-integrated learning on campus through capstone projects and CEC convenings on campus. This proposal creates *Global Futures Networks* for undergraduates by mobilizing CEE's foundational resources, including its *WaterlooWorks* platform, to support capstone/CEC WIL. CEE can also serve a secretariat role to streamline community and partner connections and communications for greater scalability across UW.

The initiative will establish on-ramps that allow students to build capstone projects from CEC courses, co-op placements and experiences with *Velocity, GreenHouse* and the *Grebel Peace Incubator* supports and resources. It also intentionally links capstone off-ramps that encourage extending student learning beyond the classroom after course completion.

CEE's central support will strengthen UW's entrepreneurial ecosystem by creating structured connections between community partners and social innovation supports like *GreenHouse, Velocity* and the *Grebel Peace Incubator*. It provides opportunities for internal and external partners to share potential capstone projects through a central platform. It will support reaching out to UW and external community partners—including government agencies, clinical settings, and industry to support CEC convenings. By partnering with *Velocity, GreenHouse*, and the *Grebel Peace Incubator*, the initiative ensures students receive essential mentorship and resources before, during and after their capstone experience. The initiative establishes clear pathways for students to develop entrepreneurial mindsets and translate academic projects into ventures through existing courses, while also providing incentives through awards for students to further develop their capstone projects after course completion. This creates a robust pipeline of innovative ideas emerging from interdisciplinary academic contexts.

This initiative reinforces UW's commitment to regional impact by creating opportunities for knowledge exchange between the university and community partners on campus. It provides engagement pathways for community organizations, municipal governments, and regional businesses to participate in various ways, such as through Capstone projects or CECs. By expanding these collaborative learning models, the initiative strengthens UW's position as an engaged anchor institution committed to local impact while addressing broader challenges.

Describe how this initiative involves collaboration across academic disciplines and organizational units at UWaterloo.

This initiative fosters collaboration across academic disciplines and organizational units at the UW by integrating and enhancing a broad spectrum of previous and ongoing efforts. It aligns with established programs such as home capstone courses, the *CEC* model, and the *i-Capstone* pilot. It also creates interdisciplinary intersections with courses like *ARTS450: Global Engagement Seminar*, *Wicked Problems*, and *WIL612: Leadership and Project Management*, enabling graduate students to be involved as instructors and mentors in capstone courses.

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Several key strategies emphasize cross-disciplinary engagement. The initiative supports a campus-wide convergence of capstone and CEC course schedules, allowing shared resources such as community partners, guest speakers, peer project reviews, and cross-course collaboration. Coordinating capstone projects across faculties enhances interdisciplinary interactions, exemplified by the upcoming collaboration between HLTH480 and BME461 in Fall 2025.

Efforts to integrate the capstone experience with existing academic structures extend further. *Map the System* interdisciplinary research projects facilitated by the *Grebel Peace Incubator* serve as both an introductory module and an essential learning component, while initiatives such as *Velocity* and *GreenHouse* provide additional support networks. The initiative also proposes adjustments to degree plans to remove barriers and encourage participation in capstone, with considerations for elective capstone opportunities as alternatives to undergraduate theses. This may be of particular interest to students who have not had the opportunity to engage in co-op (i.e., regular stream programs).

On an organizational level, leadership is encouraged to develop mechanisms for accepting cost recovery payments from private and public sector capstone partners, a practice already established in peer institutions. Additionally, aligning capstone awards, such as the *Global Futures Capstone Award*s, with other interdisciplinary awards strengthens administrative cohesion and increases organizational efficiency.

Beyond academics, the initiative emphasizes external partnerships and engagement with alums, creating a sustainable ecosystem for collaboration. Alum involvement enriches students' experiences, while external partner engagements evolve into co-op and sponsored research opportunities. A unified intellectual property framework and a coordinated approach to external communications ensure that Waterloo's capstone model stands out, leveraging the university's strengths in co-op education and interdisciplinary innovation.

Ultimately, this initiative builds a comprehensive, interconnected capstone experience that bridges faculties, institutes, students, and external partners, reinforcing the culture of interdisciplinary engagement at Waterloo.

Leadership Support Items

We are asking leadership to think differently, work together, and act with purpose in several dimensions:

1. **Support a common cross-selling "playbook" to have a One Waterloo** way of articulating co-op, sponsored grad research projects, CEC opportunities and capstone. Capstone engagement with external partners at UW differs from capstone engagement with external partners at other universities because of our world leading co-op program and unique IP policy. We all need to be on the same page about these issues, so we can communicate clearly with external partners. (Work on this playbook has already begun, in collaboration with CEE, Advancement, and Research.)
2. **Support Alumni engagement as capstone and CEC partners.** Capstone can be a powerful way for alumni to stay connected with UW and have rewarding experiences engaging with the next generation of students.
3. **Converging on a campus-wide capstone course time**, so capstone courses can share resources: e.g., guest speakers, CEC members, peer project reviews, etc. For example, in

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fall 2025 the HLTH480 and BME461 capstone courses will have overlapping times, and will be doing a number of activities together.

4. **Create capacity for UW to accept cost recovery payments from private and public sector capstone project partners.** For example, we have surveyed that Engineering faculties at peer Canadian universities have this capacity, and charge partner participation fees between $1k to $4k.
5. **Supporting peer interaction opportunities between capstone courses and first year UCR courses**. So the first year and fourth year students can give each other feedback.
6. **Supporting capstone student recruitment**. UW does not currently have a system for recommending courses to students, short of degree plans and student advisors. We propose that capstone becomes a more general part of the culture, and so is aligned with these existing recruiting systems. In addition, we propose novel new recruiting mechanisms. For example, in i-capstone we have had FCI email students at the right academic level who have had co-op jobs aligned to cities.
7. **Create a culture of encouraging students to engage with capstone project partners**. For example, 90%+ of the capstone projects in Engineering each year do not currently have a project partner, so there needs to be a gradual culture shift.
8. **Consider creating capstone course codes**, if helpful to support opportunities for students with specific degree plans. Both i-capstone and Wicked Problems have been using a handful of course codes held-with each other so that students from different majors can get a course code compatible with their degree plan. York U. also does this.
9. **Consider updating degree plans to include an elective capstone opportunity**, as appropriate. For example, Science is considering making capstone an alternative to undergraduate thesis: much of the work that could have happened in an undergraduate thesis can also happen in a capstone course, with a professor as the project partner.
10. **Coordinating the administration of the Global Futures capstone awards**. Aligning with administration of other capstone awards and related awards, such as the Linda Carson Memorial Interdisciplinary Award.
11. **See open-ended, project-oriented upper-year courses as flavours of capstone.** For example, the undergraduate experience in both *ARTS450 Global Engagement Seminar* and *Wicked Problems* is capstone-esque. See these as interdisciplinary capstone courses with subject themes. By coordinating them under the broader capstone umbrella, they get access to shared resources, external partner engagement, alignment with degree plans, and student recruitment. For Fall 2025, GES will be scheduled alongside the other capstone courses to foster interactions between students, partners, and stakeholders writ large.
12. **Support a common Intellectual Property framework for capstone**, to be developed as part of this grant, consistent with UW Policy, the laws of Ontario, etc. (Preliminary work on this has already begun, in collaboration with the UW Office of Research.)

Proposed Budget Allocation Items

1. Create a capstone instructor community of practice.
2. Create a Community-Embedded Class (CEC) community of practice.
3. Create capacity in CEE to support the transactional aspects of external partner engagement in capstone courses.

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4. Create capacity in CEE to support the transactional aspects of external partner engagement in CEC.
5. A capstone sessional instructor fund for these startup years while building enrollments.
6. Global Futures themed capstone awards.

Where applicable, describe any existing or planned internal/external partners and opportunities for sustainability of the initiative beyond the Global Futures funding.

This initiative ensures sustainability beyond the Global Futures funding by leveraging existing partnerships and cultivating new ones. Instructors have successfully drawn upon their networks to co-create capstone projects that align with both student interests and partner priorities. The cost recovery fees from capstone project partners not only support the administration of these programs through CEE but also subsidize the substantial expenses associated with project components, with engineering departments currently investing over $200,000 annually. Additionally, partnering with CEE to leverage existing assets like *WaterlooWorks* allows us to expand the capstone experience efficiently, eliminating the need for new systems and infrastructure while maximizing asset utilization.

Enrollment levels in both i-Capstone and home capstone sections remain robust, providing a strong justification for teaching task allocations. These courses, often mandatory for degree completion, maintain steady participation, reinforcing their ongoing viability. Additionally, the two communities of practice will persist either as centrally funded initiatives or as service and teaching tasks.

UW Centres and Institutes, including *FCI* and *RoboHub*, are crucial in fostering capstone partnerships. They actively engage with external partners and faculty members to identify potential projects and facilitate collaborations. Faculty members are also incentivized to propose capstone projects within the same system where external partners submit their own, enhancing engagement levels and ensuring diverse projects.

The capstone model has proven to be a successful point of interaction for external partners, some of whom evolve into co-op employers and sponsors for graduate research projects. A notable example is *Future Cities Institute*'s partnership with Iqaluit, which initially started as a capstone collaboration and has since expanded to include co-op employment opportunities.

Finally, the sustainability of the *Global Futures Capstone Awards* is not just a goal, but a reality. We have successfully developed a long-term funding model over the course of the grant period, securing the program's continuity and impact beyond the initial funding cycle. This success is a testament to the viability and impact of the initiative, and we are confident in its future sustainability.

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